

CSE 222 - Homework #8

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How to Compile?

- make: To compile the files.
- make run: To run the program.
- make doc: To create JavaDoc.

1) Person

- String name
 - int age
 - List<String> hobbies
 - Date timestamp
1. public Person(String, int, List<String>): Creates a new person with the provided data.
 2. @Override public String toString(): Returns a string including the person's name, age and hobbies.

2) Suggestion

- Person person
- double score
- int mutualFriends
- int commonHobbies

1. `public Suggestion(Person, double, int, int)`: Creates a new suggestion with the provided data.
2. `public double getScore( )`: Returns the score.

3) SocialNetworkGraph

- `Map<String, Person> people`
 - `Map<Person, List<Person>> friendships`
1. `public void addPerson(String, int, List<String>)`: Adds a new person to the network with the provided data.
 2. `public void removePerson(String)`: Removes a person with the given name from the network.
 3. `public void addFriendship(String, String)`: Makes a friendship connection between two people with the given names.
 4. `public void removeFriendship(String, String)`: Removes the friendship between two people with the given names.
 5. `public void findShortestPath(String, String)`: Finds the shortest path between two people with the given names using BFS.
 6. `public void suggestFriends(String, int)`: Generates a specified number of friend suggestions for a person.
 7. `public void countClusters( )`: Counts the clusters in the network.
 8. `private void printPath(Person, Person, Map<Person, Person>)`: Prints the provided path between two people.
 9. `private void bfs(Person, Set<Person>, List<Person>)`: Starting with the given person, generates a cluster using BFS.
 10. `private int countMutualFriends(Person, Person)`: Counts the number of mutual friends between two given people.
 11. `private int countCommonHobbies(Person, Person)`: Counts the number of

common hobbies between two given people.

4) Main

1. private static void printMenu(): Prints the CLI menu.
2. public static void main(String[]): Entry point of the program. Takes input from the user and makes operations on the network.

Example Outputs

```
PS C:\Workspace\2023-spr\cse222\hw08\HW8_Files> make run
java Main
Person added: zeynep (Age: 25, Hobbies: [reading, hiking, cooking]) Timestamp: Tue May 28 21:28:23 TRT 2024
Person added: duru (Age: 22, Hobbies: [swimming, cooking]) Timestamp: Tue May 28 21:28:23 TRT 2024
Person added: talha (Age: 27, Hobbies: [hiking, painting]) Timestamp: Tue May 28 21:28:23 TRT 2024
Person added: ahmet (Age: 30, Hobbies: [reading, swimming]) Timestamp: Tue May 28 21:28:23 TRT 2024
Person added: arda (Age: 28, Hobbies: [running, swimming]) Timestamp: Tue May 28 21:28:23 TRT 2024
Person added: Frank Wilson (Age: 26, Hobbies: [reading, hiking]) Timestamp: Tue May 28 21:28:23 TRT 2024
Friendship added between zeynep and duru
Friendship added between zeynep and talha
Friendship added between duru and arda
Friendship added between talha and ahmet
Friendship added between ahmet and arda
One or both person not found in the network!
Cluster 1:
ahmet
talha
arda
zeynep
duru
Cluster 2:
Frank Wilson
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: █
```

Note: Default values added for demonstration purposes.

```
Please select an option: 1
Enter name & surname: test test
Enter age: 34
Enter two hobbies
-----
1 : hobby1
2 : hobby2
Person added: test test (Age: 34, Hobbies: [hobby1, hobby2]) Timestamp: Tue May 28 21:31:19 TRT 2024
===== Social Network Analysis Menu =====
1. Add person
```

```
=====
Please select an option: 7
Cluster 1:
test test
Cluster 2:
ahmet
talha
arda
zeynep
duru
Cluster 3:
Frank Wilson
```

```
Please select an option: 3
First person name: Frank Wilson
Second person name: test test
Friendship added between Frank Wilson and test test
===== Social Network Analysis Menu =====
```

```
Please select an option: 7
Cluster 1:
test test
Frank Wilson
Cluster 2:
ahmet
talha
arda
zeynep
duru
```

```
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 2
Person name to be removed: deneme
Person not found!
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 2
Person name to be removed: test test
Person removed: test test (Age: 34, Hobbies: [hobby1, hobby2])
```

```
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 4
First person name: test test
Second person name: Frank Wilson
One or both persons not found in the network.
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 7
Cluster 1:
ahmet
talha
arda
zeynep
duru
Cluster 2:
Frank Wilson
```

```
==== Social Network Analysis Menu ====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 5
Start name: zeynep
End name: arda
Shortest path: zeynep duru arda
```

```
==== Social Network Analysis Menu ====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 4
First person name: zeynep
Second person name: duru
Friendship removed between zeynep and duru
```

```
==== Social Network Analysis Menu ====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 5
Start name: zeynep
End name: arda
Shortest path: zeynep talha ahmet arda
```

```
==== Social Network Analysis Menu ====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 6
Name: zeynep
Maximum # of friend to suggest: 4
Suggested friends for zeynep:
ahmet (Score: 3,2, 1 mutual friends, 1 common hobbies)
Frank Wilson (Score: 2,4, 0 mutual friends, 2 common hobbies)
duru (Score: 1,2, 0 mutual friends, 1 common hobbies)
arda (Score: 0,0, 0 mutual friends, 0 common hobbies)
```

```
===== Social Network Analysis Menu =====
1. Add person
2. Remove person
3. Add friendship
4. Remove friendship
5. Find shortest path
6. Suggest friends
7. Count clusters
8. Exit
Please select an option: 6
Name: zeynep
Maximum # of friend to suggest: 4
Suggested friends for zeynep:
ahmet (Score: 3,2, 1 mutual friends, 1 common hobbies)
Frank Wilson (Score: 2,4, 0 mutual friends, 2 common hobbies)
duru (Score: 1,2, 0 mutual friends, 1 common hobbies)
arda (Score: 0,0, 0 mutual friends, 0 common hobbies)
```